

Daily Logic Drill #3

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Negating Statements · Negating Quantifiers ($\forall/\exists$) · Contrapositive · Proof by Contrapositive.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Negate a statement, or state a contrapositive. Pure recognition.

1. Negate: “ n is even.”
2. Negate: “ $x > 5$.”
3. Negate: “ n is divisible by 3 and by 5.”
4. Negate: “ $x = 2$ or $x = 3$.”
5. Negate: “For all integers n , $n^2 \geq 0$.”
6. Negate: “There exists an integer n such that $n^2 = 2$.”
7. True or false: “The negation of a true statement is always false.”
8. State the contrapositive of: “If n is even, then n^2 is even.”
9. True or false: “The contrapositive of a statement is always logically equivalent to the original.”
10. True or false: “The contrapositive of a statement is the same as its converse.”

Section B Manipulation Drills

(≤50 s each)

Negate an implication or a compound statement; build your first contrapositive proof.

1. Negate: “If n is prime, then n is odd.”
2. Negate: “For all $x > 0$, $x^2 > 0$.”
3. Negate: “ n is a multiple of 4 or n is a multiple of 6.”
4. State the contrapositive of: “If $x > 2$, then $x^2 > 4$.”

5. Prove “If n^2 is odd, then n is odd” by proving its contrapositive.
6. Negate: “ n is prime and $n > 2$.”
7. Negate: “If it rains, then the ground is wet.”
8. State the contrapositive of: “If n is a multiple of 9, then n is a multiple of 3.”
9. True or false: “ $\text{not}(P \text{ and } Q)$ ” is the same as “ $(\text{not } P) \text{ and } (\text{not } Q)$ ”.
10. True or false: “ $\text{not}(P \text{ or } Q)$ ” is the same as “ $(\text{not } P) \text{ and } (\text{not } Q)$ ”.

Section C Substitution & Structure**(≤ 90 s each)**

Negate quantified statements, prove via contrapositive, and spot a De Morgan trap.

1. Negate: “For every prime p , p is odd or $p = 2$.”
2. Prove “If n^2 is even, then n is even” via its contrapositive.
3. Which is the correct negation of “If x is rational, then x^2 is rational”?
 - A) If x is irrational, then x^2 is irrational.
 - B) x is rational and x^2 is irrational.
 - C) If x^2 is irrational, then x is irrational.
 - D) x is irrational and x^2 is rational.
4. Negate: “There exists an integer n such that n is prime and n is even.”
5. State and prove the contrapositive of: “If n is a multiple of 4, then n^2 is a multiple of 16.”
6. True or false, with justification: “If a statement’s contrapositive is false, the original statement must also be false.”
7. Negate: “There exists a positive integer n such that n is not a multiple of any prime less than 10.”
8. Prove “If n is not a multiple of 3, then n^2 is not a multiple of 3” via its contrapositive.

Section D Challenge**(TMUA / SMC difficulty)**

Insight over computation. Each question has a short, elegant solution — find it.

1. Given the statement S : “If n is a multiple of 4, then n is even.” Which of the following is the contrapositive of S ?
 - A) If n is not even, then n is not a multiple of 4.
 - B) If n is even, then n is a multiple of 4.

C) If n is not a multiple of 4, then n is not even.

D) If n is odd, then n is a multiple of 4.

2. Prove by contrapositive: “If $n^2 - 2n$ is odd, then n is odd.”

3. On an island, every inhabitant either always tells the truth or always lies. You meet two people, A and B . A says: “At least one of us is a liar.” Determine, with justification, whether A and B are truth-tellers or liars.

4. Determine, with proof via contrapositive, whether the following is true: “If n^3 is even, then n is even.”

5. Statement S : “If n is a multiple of 15, then n is a multiple of 3 and a multiple of 5.” (a) State the contrapositive of S . (b) Is “ n is a multiple of 15” necessary and sufficient for “ n is a multiple of 3 and a multiple of 5”? (c) Negate S directly (without first simplifying via contrapositive).

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths